Heart Disease Risk Prediction- Bayesian Modeling and MLP

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DS 4420 Final Project Poster

Northeastern University

Introduction/Abstract

- This project applies machine learning to improve **Coronary heart disease (CHD)** risk prediction using the famous Framingham Heart Study dataset. In this project, we compare two modeling approaches—**Bayesian logistic regression** and **Multilayer Perceptron (MLP)**—to explore the trade-offs between interpretability and predictive power in a medical context.
- Our findings show how probabilistic models can offer uncertainty estimates essential in clinical decision-making, while neural networks may offer higher accuracy when optimized correctly. *Ultimately, our goal is to improve early intervention through accessible, data-driven risk assessment tools.*

Motivation and Data

Risk prediction plays a critical role in preventative healthcare by identifying individuals at higher risk and enabling them to adopt positive lifestyle choices, or seek medical help before irreversible damage occurs. Traditionally, heart disease is diagnosed through clinical methods such as patient history, physical examinations with a stethoscope, and diagnostic tools like ultrasound and electrocardiograms (ECG). However, limitations such as human error, lack of equal access to adequate medical infrastructure, and delays in receiving results, highlight the need for supplementary, data-driven diagnostic tools. Predictive modeling, while not a substitute for clinical diagnosis, offers a first step line assessment tool to identify possible at-risk individuals and support early intervention strategies.

Therefore, to address the binary classification task of predicting CHD, we applied two machine learning methods:

- **Bayesian Logistic Regression in R**
- **Multilayer Perceptron (MLP) Neural Network in Python**

The dataset we used in this study is derived from the **Framingham Heart Study**, a long-term, ongoing cardiovascular cohort study of about 5,000 subjects initiated in 1948 in Framingham, Massachusetts. The dataset includes age, sex, total cholesterol, glucose, blood pressure, BMI, smoking status, and treatment history—factors that are both clinically relevant and statistically informative for CHD risk modeling.

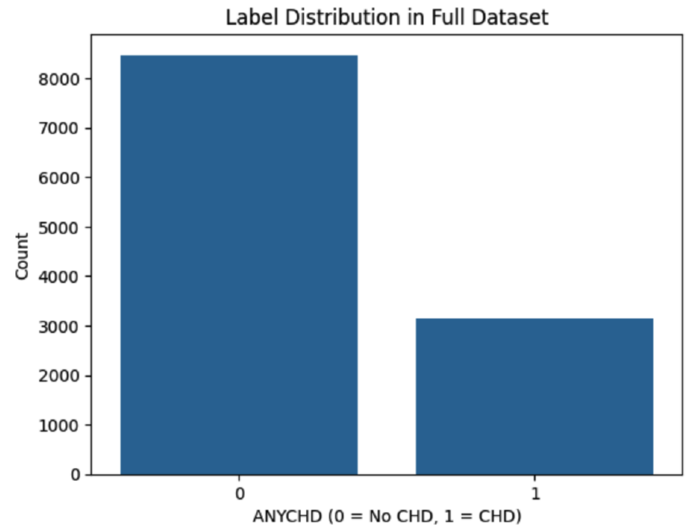

Fig. 1: Class distribution of 0's vs 1's for CHD (target variable)

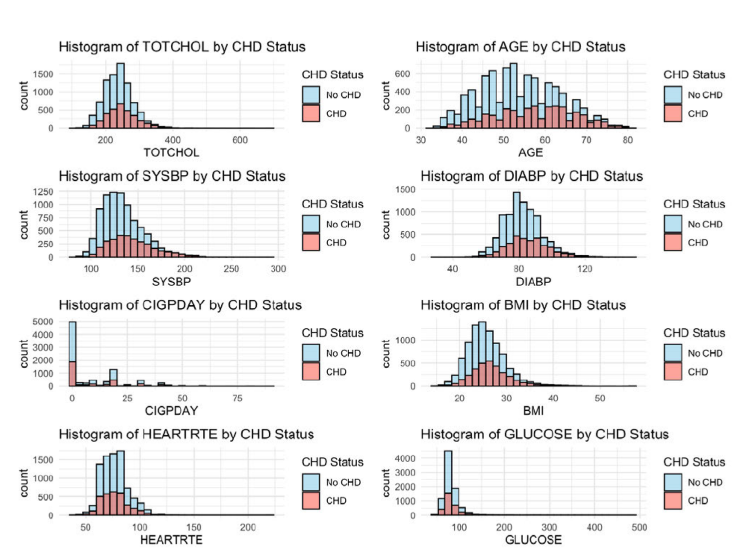

Fig. 3: Histogram of selected features in data

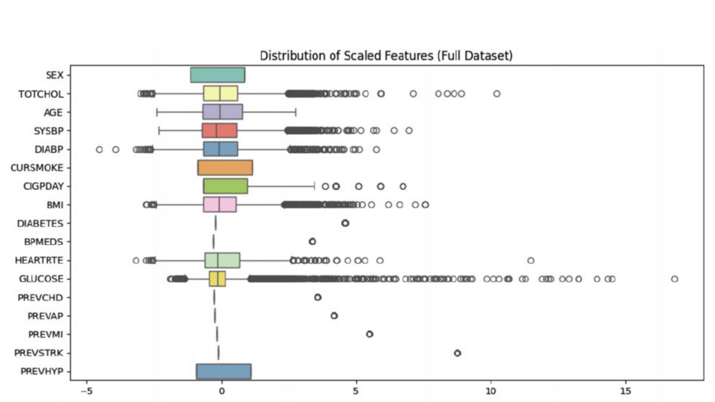

Fig. 2: Boxplot of all features in data (scaled)

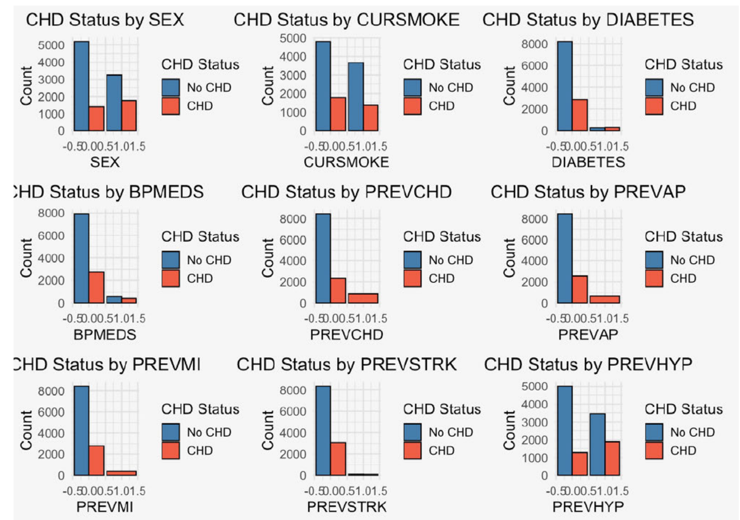

Fig. 4: Bar Graphs of selected features in data

Methodology

Cleaning: To ensure data quality and relevance, we implemented comprehensive cleaning and pre-processing. Columns with more than 50 percent missing values were removed. Then missing values in continuous variables were imputed using the median, while binary categorical variables were imputed using the mode. Logical imputation was applied, and we removed "leakage" variables—features like DEATH, MI, and TIMECHD that reflect future outcomes since these could artificially inflate model performance. The final cleaned dataset included 11,627 rows, and 18 columns.

R, Bayesian Modeling: First, we implemented a Bayesian logistic regression model using the brms package in R to predict the probability of developing coronary heart disease (CHD). We specified informative priors for each of the predictor variables based on existing medical literature, including sources such as BMJ Global Health, the CDC, and the New England Journal of Medicine. Then, the Bayesian model was trained using four MCMC chains with 2,000 iterations each and a warmup of 360 iterations (18 percent). We chose the Bernoulli likelihood with a logit link function appropriate for our binary outcome variable (ANYCHD).

Python, MLP: As our second modeling approach, we implemented a Multilayer Perceptron (MLP). The network consisted of an input layer aligned to our 18 predictor variables, a single hidden layer with 32 neurons activated by the ReLU function, and an output layer using a Sigmoid activation function to generate a probability for the binary outcome variable (ANYCHD). We initialized the hidden layer's parameters using He initialization to ensure stable gradient flow through the ReLU-activated hidden layer. The model used binary cross-entropy loss, a standard choice for binary classification tasks. Training was then conducted over 600 epochs using a learning rate of 0.01 and a batch size of 32.

Results

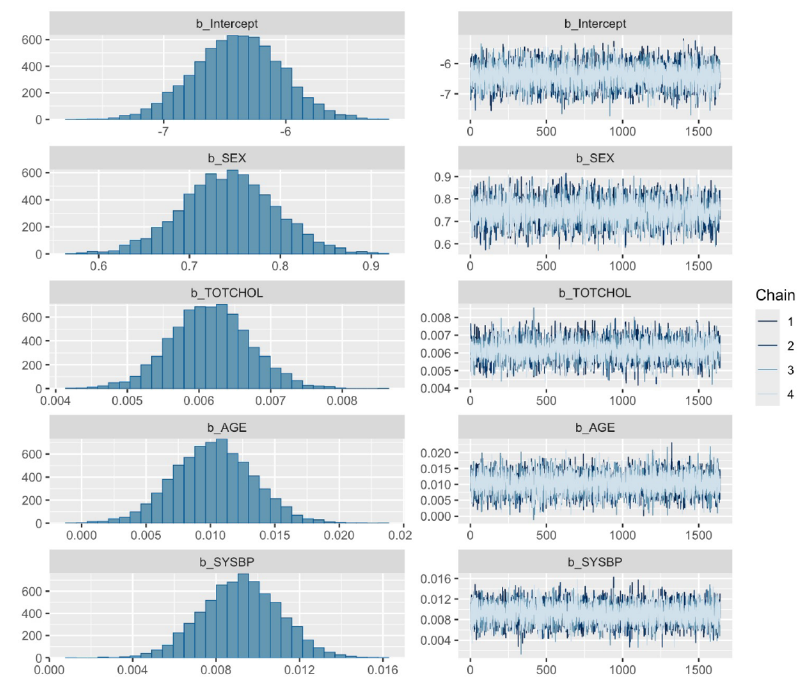

Fig. 5: Bayesian Model Results

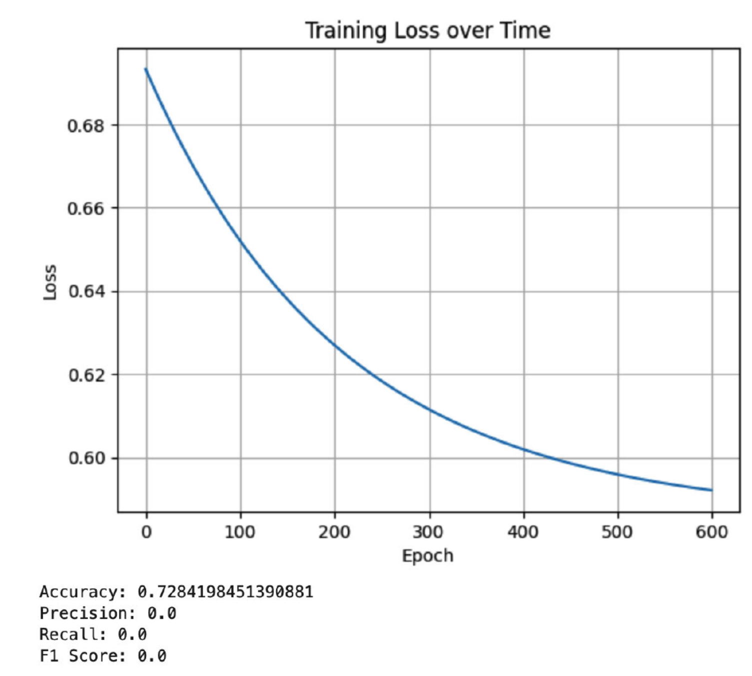

Fig. 6: MLP Training Loss and Accuracy on Test set

Bayesian Model:

- Accuracy: 83.6%, Precision: 0.696, Recall: 0.241, F1 Score: 0.359
- Bayesian R^2: 0.153 — the model explains 26.5% of variance
- Significant positive predictors: age, cholesterol, BP, BMI, smoking, diabetes, male sex, prior CHD
- Negative predictor: heart rate (protective effect)
- Strong interpretability via posterior distributions

MLP Model:

- Accuracy: 72.84%, but failed to predict ANYCHD=1 due to class imbalance
- Precision, Recall, and F1: 0 (model collapsed to majority class)

Discussion / Future Work / Web App

This project demonstrates the complementary strengths of Bayesian and neural network approaches in predicting CHD risk. The Bayesian model was interpretable and uncertainty-aware, while the MLP offered greater flexibility—but struggled due to class imbalance.
In future iterations, we aim to:

- Address class imbalance with SMOTE or class-weighted loss
- Optimize model thresholds to improve recall
- Evaluate external validity using cross-validation or new datasets
- Integrate SHAP or LIME for MLP interpretability

We also deployed our Bayesian model via a Shiny web app that allows users to input risk factors and visualize personalized predictions with uncertainty intervals.

Scan to try the app for yourself!

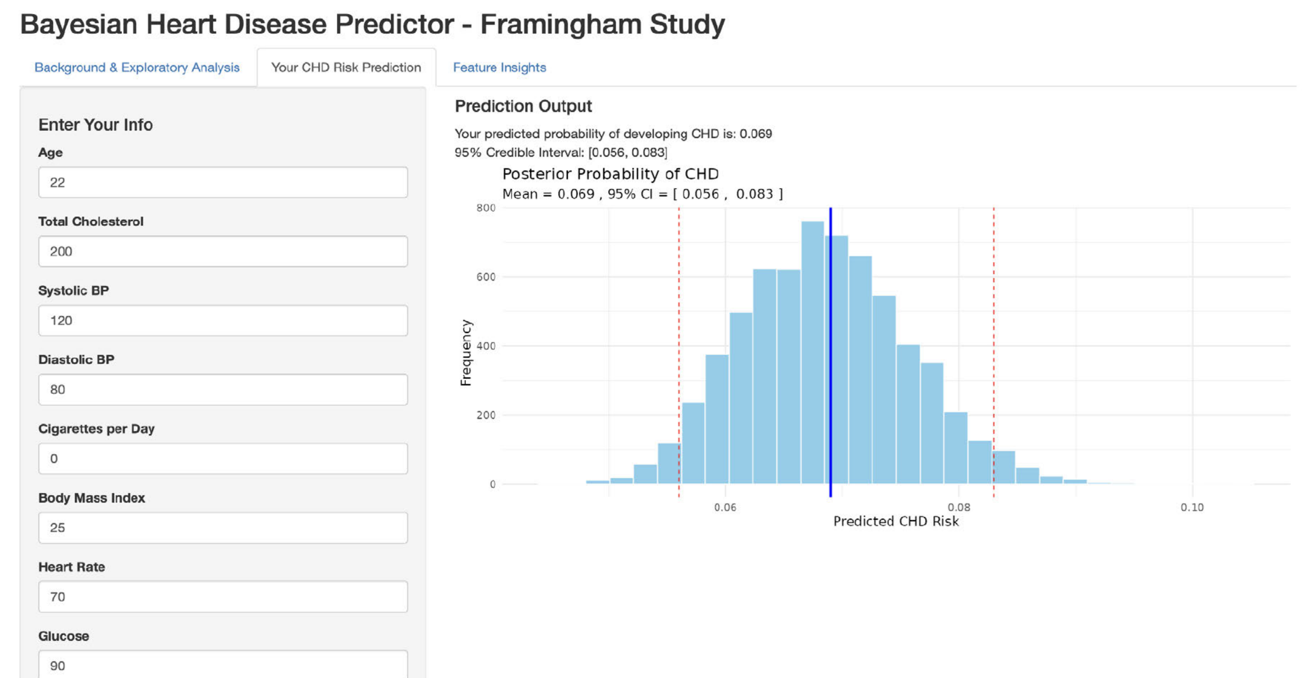

Fig. 7: App Patient Prediction Output

References

1. National Heart, Lung, and Blood Institute. (n.d.). Framingham Heart Study (FHS). U.S. Department of Health and Human Services. https://www.nhlbi.nih.gov/science/framingham-heart-study-fhs
2. American Heart Association News. (2021, February 1). More than half of U.S. adults don't know heart disease is leading cause of death, despite 100-year reign. American Heart Association Newsroom. https://newsroom.heart.org/news/more-than-half-of-u-s-adults-dont-know-heart-disease-is-leading-cause-of-death-despite-100-year-reign

Find an extensive list in our official report!